

ISYE 6740 Project Proposal

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1 Introduction and Problem Statement

Measurement of risk in finance has evolved significantly over the years, but a staple metric in use has been Value-at-Risk or VaR. VaR addresses a critical question: *What is the minimum amount that I can expect to lose with a certain probability over a given horizon?*^[1]

There are multiple methods used to calculate VaR, but one of the most simple would be the historical method, which takes a historical set of time series returns, approximates them with a normal distribution, and takes the loss corresponding to a particular confidence interval, e.g., the 95% confidence interval as the VaR estimate.

Another popular method is Monte-Carlo VaR where thousands of scenarios are instead randomly generated based on historical returns, e.g., 5000 times and a similar approach of referencing the loss at a chosen confidence interval is used to calculate VaR.

However, there are some issues with these contemporary methods of calculating VaR. Financial markets tend to underreact to minor events and overreact to major ones. Past studies also indicate that returns also exhibit asymmetry, reacting slightly stronger to positive than negative events^[2]. Consequently, the widely-used assumption that returns are normally distributed does not hold in reality.

This dichotomy has given rise to other types of metrics, such as Expected Shortfall, which gives an estimate to the expected losses beyond the VaR threshold. In recent years, the advent of Deep learning models offers a new opportunity to refine VaR calculation methodologies. Techniques such as Long short-term memory (LSTM) recurrent neural networks^[3] could potentially capture the non-linear nature of stock returns more accurately, potentially providing a more robust approach to financial risk measurement.

2 Proposed Methodology

We will utilize a ten-year historical period of daily returns from a universe of US equities drawn entirely from the S&P 500 index. These returns will be calculated using adjusted closing prices obtained from publicly available data sources such as the Yahoo Finance API package in Python. The below diagram shows a distribution of Microsoft stock returns across the past 10 years. Notice the peaked center and fat tails, indicative of a leptokurtic distribution with a higher than normal probability of extreme events.

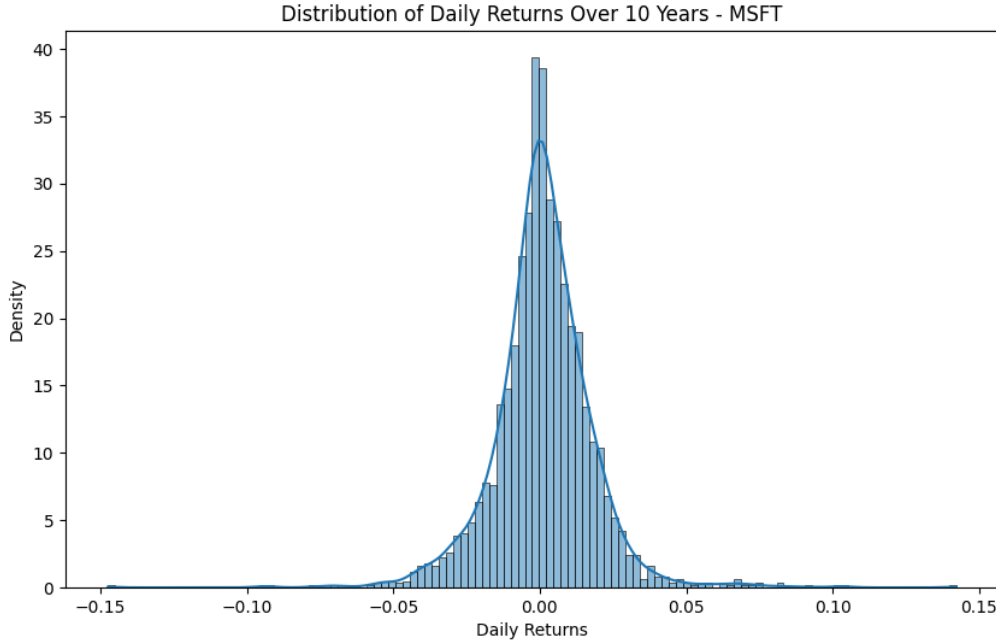


Figure 1: Histogram showing the distribution of a ten-year period of daily historical returns of a sample US equity, i.e., Microsoft (MSFT US Equity) stock.

We can then build an LSTM Recurrent Neural Network (RNN) model and train it on the historical returns data to capture the temporal dependencies inherent in financial time series data. The model will then be trained to forecast the next daily return value based on past observations.

The trained LSTM model will then be used to forecast 1000 future time series scenarios. For each scenario, we will calculate the corresponding profit or loss value. Using these simulated returns, we can then proceed to calculate VaR at specified confidence intervals, such as 95% and 99%. Additionally, we will also calculate VaR using extant methodologies such as a Monte-Carlo VaR with 1000 simulations, and historical VaR using the ten-year historical return dataset.

All VaR calculations will be based on daily returns with a 1-year lookback period by default.

2.1 Portfolio-based measurements

While it might be insightful to calculate individual stock-level VaR values, it makes more sense in practice to analyze these metrics at the portfolio level. We will thus generate multiple portfolios based on the existing universe of stocks in our training data. The portfolios will be constructed arbitrarily by using Principal Components Analysis (PCA) on the historical returns of the universe to obtain the key principal components.

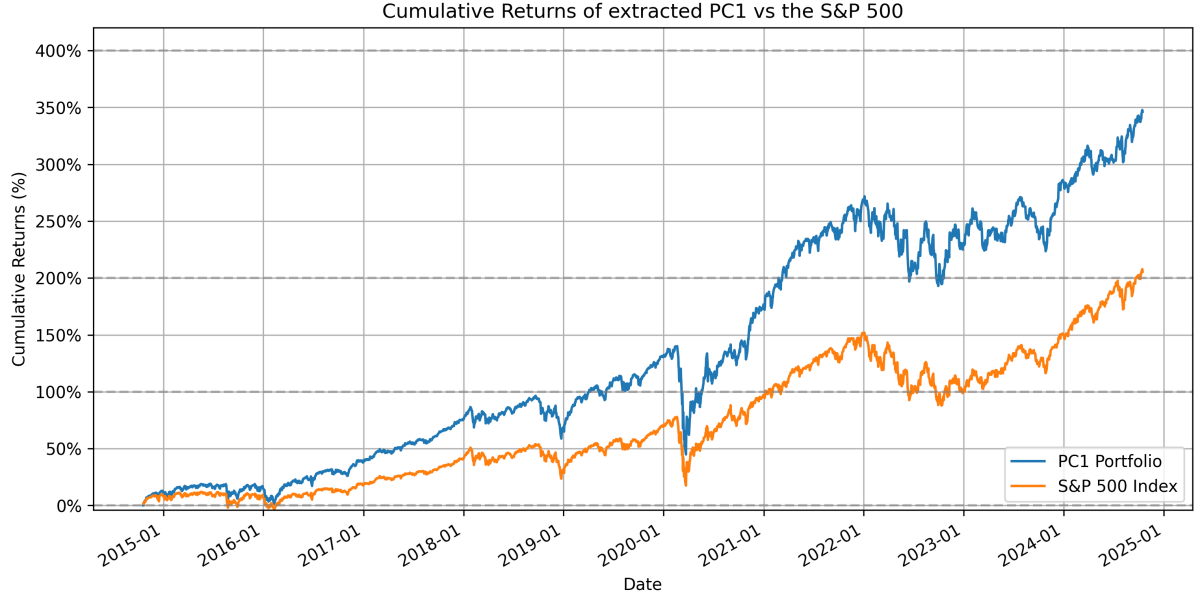


Figure 2: Cumulative Portfolio Returns Over Time for a simulated Portfolio using the first extracted principal component compared to the S&P 500 over a ten-year historical period.

These principal components will then be converted into suitable factor loadings, which will serve as portfolio weights[4] that can then be used to construct multiple sample portfolios for use in our VaR evaluation and comparison strategies. The above diagram is a simple example of how a portfolio made from first principal component extracted from individual S&P 500 equities outperforms the said index's performance over the past 10 years.

3 Planned Evaluation Strategies

The methodologies proposed to calculate VaR will then be applied to these simulated portfolios, and we will calculate VaR across a rolling 1-year lookback period based on the extracted raw data. The following strategies will be implemented to compare the effectiveness of the different VaR methodologies.

3.1 Tests of Normality

To determine whether or not our portfolio returns indeed follow a normal distribution, we can create histograms and compare the return distributions for the P%Ls generated in all three scenarios; LSTM, Monte-Carlo, and Historical. Additionally, Q-Q plots can be utilized to visually assess the normality of the simulated return distributions.

3.2 Degree of Conservativity

We can also look into confirming the conservativeness of the VaR estimates by examining the magnitude of the VaR values. A higher VaR indicates a less conservative estimate, as it suggests that larger portfolio losses are required to breach the VaR threshold.

3.3 VaR Backtesting

To determine which VaR metric is better, we will conduct historical backtesting. This will involve saving the most recent 6-months of return data and comparing them to the 1-year rolling VaR values calculated for each method (LSTM, Monte Carlo, Historical VaR). We can

then track the number of times the VaR threshold is hit by comparing the daily portfolio returns to the computed VaR levels across the six-month period. This can be used to calculate a "hit-rate" metric, where fewer hits (breaches) would indicate a more robust and accurate VaR model.

3.4 Responsiveness during periods of High Volatility

To assess the responsiveness of these VaR thresholds during market stress, we can also focus on periods of heightened volatility, as indicated by metrics like the VIX Index. Specifically, we can analyze periods of heightened volatility, such as the Covid-19 financial crisis from February-March 2020. Comparing the frequency and magnitude of VaR breaches during these stressed periods could also prove insightful in proving whether our LSTM model better captures non-linear trends and perform more effectively than contemporary VaR models.

References

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